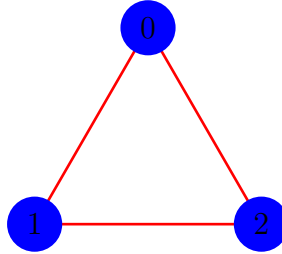


Homework 5

1. Consider a triangular molecule (equilateral):

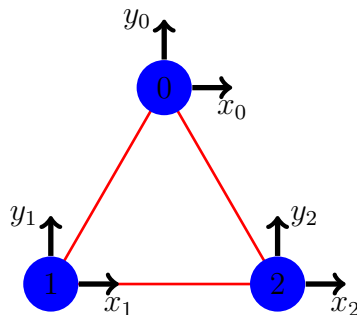


For this problem we will assume that the molecule can only move out of the page (z-direction). If this molecule is not anchored to anything, one can prove that at second order it has a potential of the form:

$$V(z_0, z_1, z_2) = \alpha(2z_0 - z_1 - z_2)^2 + 3\alpha(z_1 - z_2)^2$$

- (a) Construct the stiffness matrix $\hat{\mathbf{V}}$ (ie. the equation of motion) in the usual naive basis of $|Z_0\rangle = [1 \ 0 \ 0]^\dagger$, etc. You should find a full 3x3 matrix.
 - (b) Compute the Eigenvalues and Eigenvectors. Please make note of any degeneracy.
2. Now let us bring some group theory to this problem.
- (a) Find all of the group elements that leave the above triangle invariant (ie. point group).
 - (b) Verify that all these alleged group elements commute with stiffness matrix from previous problem.
 - (c) Construct the multiplication table, and verify that a group is formed.
 - (d) Construct the classes.
 - (e) Derive the character table. It is not hard to do in this case.
 - (f) Construct a representation of the three atoms (ie. a 3x3 matrix for each element of the group).
 - (g) Decompose this representation into irreducible representations. Which ones are present?
 - (h) Use the projection operator to construct the irreducible basis vectors. Use these to block diagonalize the stiffness matrix from the previous problem.
 - (i) Using the multidimensional symmetrized modes, construct the multidimensional irreducible representation matrices.

- (j) Now consider the possibility that the atoms can only move in the x-y plane, as shown below. Construct a representation using the six basis vectors (ie. a 6x6 matrix for each element of the point group).



- (k) Decompose this representation into irreducible representations. Which ones are present?
- (l) Use the projection operator to construct the irreducible basis vectors.